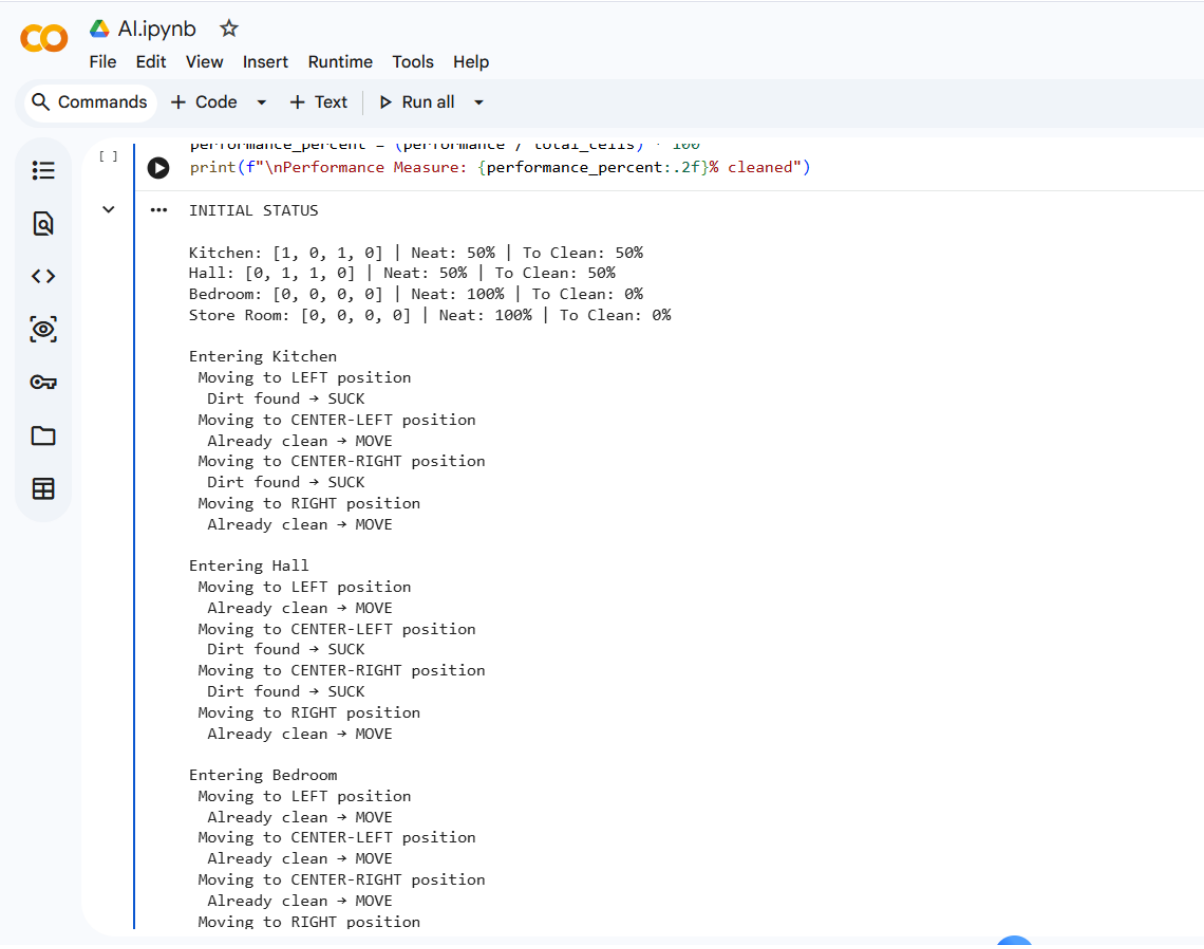
Vacuum Cleaner 2:

```
rooms = {
    "Kitchen":    [1, 0, 1, 0],
    "Hall":       [0, 1, 1, 0],
    "Bedroom":    [0, 0, 0, 0],
    "Store Room": [0, 0, 0, 0]
}

performance = 0
total_cells = 0

# Count total cells
for room in rooms.values():
    total_cells += len(room)

# Position names
positions = {
    0: "LEFT",
    1: "CENTER-LEFT",
    2: "CENTER-RIGHT",
    3: "RIGHT"
}

def vacuum_agent(room_name, room):
    global performance
    print(f"\nEntering {room_name}")

    for i in range(len(room)):
```

```
        print(f" Moving to {positions[i]} position")

        if room[i] == 1:
            print(f"  Dirt found → SUCK")
            room[i] = 0
            performance += 1
        else:
            print(f"  Already clean → MOVE")

def room_status(room):
    clean = room.count(0)
    dirty = room.count(1)
    neat_percent = (clean / len(room)) * 100
    dirty_percent = (dirty / len(room)) * 100
    return neat_percent, dirty_percent

# INITIAL STATUS
print("INITIAL STATUS\n")
for name, room in rooms.items():
    neat, dirty = room_status(room)
    print(f"{name}: {room} | Neat: {neat:.0f}% | To Clean: {dirty:.0f}%")

# AGENT ACTION
for name, room in rooms.items():
    vacuum_agent(name, room)

# FINAL STATUS
print("\nFINAL STATUS\n")
```

```
for name, room in rooms.items():
    neat, dirty = room_status(room)
    print(f"{name}: {room} | Neat: {neat:.0f}% | To Clean: {dirty:.0f}%")

# PERFORMANCE MEASURE
performance_percent = (performance / total_cells) * 100
print(f"\nPerformance Measure: {performance_percent:.2f}% cleaned")
```

<u>output:</u>



AI.ipynb

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```
Entering Bedroom
 Moving to LEFT position
  Already clean → MOVE
 Moving to CENTER-LEFT position
  Already clean → MOVE
 Moving to CENTER-RIGHT position
  Already clean → MOVE
 Moving to RIGHT position
  Already clean → MOVE

Entering Store Room
 Moving to LEFT position
  Already clean → MOVE
 Moving to CENTER-LEFT position
  Already clean → MOVE
 Moving to CENTER-RIGHT position
  Already clean → MOVE
 Moving to RIGHT position
  Already clean → MOVE

FINAL STATUS

Kitchen: [0, 0, 0, 0] | Neat: 100% | To Clean: 0%
Hall: [0, 0, 0, 0] | Neat: 100% | To Clean: 0%
Bedroom: [0, 0, 0, 0] | Neat: 100% | To Clean: 0%
Store Room: [0, 0, 0, 0] | Neat: 100% | To Clean: 0%

Performance Measure: 25.00% cleaned
```

```
from collections import deque

def bfs_shortest_path(graph, start):
    visited = [False] * len(graph)
    distance = [-1] * len(graph)

    queue = deque()
```

Variables Terminal

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